Vectors

1 Vector Equation of a Line 3
1.1 The Different Forms of Equations 3
1.2 Finding Vector Equations and Points on Lines . . 6
1.3 Multiple Lines 7

2 Mathematics 11
2.1 Theorems 11
2.2 Definitions 11
2.3 Notations 11
2.4 Remarks 12
2.5 Corollaries 12
2.6 Propositions 12
2.7 Examples 12
2.8 Exercises 12
2.9 Problems 13
2.10 Vocabulary 13

3 Presenting Information and Results with a Long Chapter Title 15
3.1 Table 15
3.2 Figure 15

1. Vector Equation of a Line

1.1 The Different Forms of Equations

1.1.1 The Vector Form

We have already met the slope-intercept equation of a line, $y = mx + c$. It turns out that we can equivalently use vectors to represent lines, using the following equation:

$$\mathbf{p} = \mathbf{a} + \lambda \mathbf{d} \tag{1.1}$$

I will now try to give some intuition about the equation. Consider the line drawn in Figure 1.1. The line goes through two points, A and B, with position vectors $\mathbf{a}$ and $\mathbf{b}$. We need to find a succint equation which represents all of the points on this line.

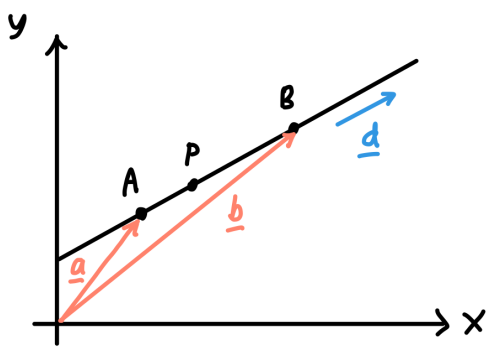

Figure 1.1: A line in 2-dimensions

If we start at A, and move to B, we have traversed the line segment AB, which can be described using the displacement vector $\overrightarrow{AB}$. This is called a direction vector: any vector which is parallel to the line ($\overrightarrow{BA}$ would also be a direction vector). We write our chosen direction vector as $\mathbf{d}$. Hopefully, you can see that for any general point P on the line, with position vector $\mathbf{p}$, $\overrightarrow{AP}$ must be parallel to $\mathbf{d}$. Recall that any two parallel vectors must be scalar multiples of each other:

$$\overrightarrow{AP} = \mathbf{p} - \mathbf{a} = \lambda \mathbf{d} \implies \mathbf{p} = \mathbf{a} + \lambda \mathbf{d}$$

where λ is a scalar. This brings us to the final form of the vector equation of a line. The position vector of any point on the line $\mathbf{p}$ can be represented using the position vector of a point which is known to be on the line $\mathbf{a}$ and the direction vector of the line $\mathbf{d}$.

R For a visualization of this, visit the following 3D Desmos graph: https://www.desmos.com/3d/x11krwktpa

Worked Example Find the vector equation of the line passing through $A(4,6)$ with direction $\mathbf{i}+2\mathbf{j}$.

As we showed above, in order to write down the vector equation of a line, we need the coordinates of a point on that line and a direction vector for that line. This question has given us both of those directly. Hence, we can write:

$$\mathbf{a} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}$$
$$\mathbf{d} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

From here, we can simply write down the equation of the line:

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Here, x and y are the coordinates of point P as described above

1.1.2 The Parametric Form

We can write the **parametric form** of a line by considering the components of the vector form:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} + \lambda \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

For the parametric form, x, y, and z are the coordinates of a general point on the line (P, in the above), x_1, y_1, and z_1 are the coordinates of a known point on the line (A, in the above), and d_1, d_2, and d_3 are the components of the direction vector ($\mathbf{d}$, in the above).

The final parametric form is given by separating the components into individual equations:

$$x = x_1 + \lambda d_1$$
$$y = y_1 + \lambda d_2$$
$$z = z_1 + \lambda d_3$$

The parametric form can then be used to find the Cartesian form.

1.1.3 The Cartesian Form

The **Cartesian form** is obtained by eliminating λ from the equations in the parametric form. First, we rearrange both equations in terms of λ:

$$\lambda = \frac{x - x_1}{d_1}$$
$$\lambda = \frac{y - y_1}{d_2}$$
$$\lambda = \frac{z - z_1}{d_3}$$

We can then simply write the Cartesian form:

$$\frac{x - x_1}{d_1} = \frac{y - y_1}{d_2} = \frac{z - z_1}{d_3} \tag{1.2}$$

R The denominators of each of the terms in the Cartesian form make up the direction vector of the line — this is a handy way of finding direction vectors quickly if you're given the equation in Cartesian form

If any of the components of the direction vector are zero, we have to write the Cartesian form as two (or more) separate equations, as below:

- If only one component of **d** is zero (e.g. d_1), we write the Cartesian form as:

$$x = x_1, \frac{y - y_1}{d_2} = \frac{z - z_1}{d_3}$$

- If two components are zero (e.g. d_1 and d_2), we write the Cartesian form as:

$$x = x_1, y = y_1, \frac{z - z_1}{d_3} = \lambda$$

1.1.4 Exercises

1.2 Finding Vector Equations and Points on Lines

1.2.1 Finding Vector Equations

You may be asked to find the vector equation of a line. In this case, you will be given either:

(a) the coordinates of a point which the line passes through and the direction of the line
(b) the coordinates of two points which the line passes through

The former is fairly easy to deal with, as the coordinates of the point are the components of $\mathbf{a}$, and the direction vector is equal to $\mathbf{d}$.

In the case that you are given two points, you can use the coordinates of either of these points as the components of $\mathbf{a}$. However, you will need to determine $\mathbf{d}$ yourself. You can do this by simply determining either displacement vector between the two points ($\overrightarrow{AB}$ or $\overrightarrow{BA}$). This vector is bound to be parallel to the line (if you don't believe this, look at $\overrightarrow{AB}$ in Figure 1.1), and can hence be used as $\mathbf{d}$.

1.2.2 Normal Lines

You may sometimes be asked to determine the equation of a line which is normal (i.e. perpendicular) with another line. If the direction vector of a line is given by $\mathbf{d} = d_1\mathbf{i} + d_2\mathbf{j}$, then one possible normal vector is given by: $\mathbf{n} = d_2\mathbf{i} - d_1\mathbf{j}$ (if you don't believe this, try finding the dot product of the two vectors).

Remember, there are an infinite number of possible normal vectors: any vector parallel to $\mathbf{n}$ is also a normal vector.

1.2.3 Checking if a Point is On A Line

To check if a point, R (position vector $\mathbf{r}$), lies on a line defined by the vector equation of a line, we simply sub $\mathbf{r}$ into $\mathbf{p}$ and check if there exists some value of λ for which the equation is satisfied:

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$$

This is best solved by breaking down the vectors into their individual components, then solving for λ individually. If a coherent value of λ exists, the point lies on the line. Otherwise, it does not.

1.2.4 Exercises

1.3 Multiple Lines

1.3.1 Writing Equations

When we write the equations for multiple lines, we need to be able to differentiate between the various lines that we have modelled. To this extent, we name each line using the L_n nomenclature. We must also remember that our scalar multiple (λ, in our case) needs to be changed for different lines.

1.3.2 Relationships between Lines

In a 2D plane, any two lines will follow one of these relations:

(a) the lines are parellel
(b) the lines are intersecting
(c) the lines are coincident (they are the same line)

For two lines to be **intersecting**, they must have a point of intersection (i.e. there must exist at least one point that satisfies both line equations). For two lines to be **parallel**: their direction vectors must be scalar multiples of one another. For two lines to be **coincident**, they must be both parallel *and* intersecting.

In 3 dimensions, there is an additional possibility: lines can be **skew**. This means that the lines are neither parallel nor intersecting (this is not possible in 2 dimensions, as any two lines that aren't parallel must eventually run into each other). To determine whether two lines are skew, we simply check that they are not parallel and not intersecting.

1.3.3 Finding Points of Intersection

Assuming we have the equations of two lines:

$$L_1 : \mathbf{p} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$
$$L_2 : \mathbf{p} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} + \mu \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$$

Note that in the above, line L_1 has direction vector $d_1\mathbf{i} + d_2\mathbf{j} + d_3\mathbf{k}$ and line L_2 has direction vector $v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}$

To find a point of intersection between the two lines, we must equate the equations (just as you would in slope-intercept form). We then separate the vector equations into three equations, one for each component:

$$a_1 + \lambda d_1 = b_1 + \mu v_1$$
$$a_2 + \lambda d_2 = b_2 + \mu v_2$$
$$a_3 + \lambda d_3 = b_3 + \mu v_3$$

This is a system of equations: solve any two to find the values of λ and μ, then substitute those values back into the third equation to see if they work.

Worked Example Find the point of intersection between lines L_1 and L_2 with equations:

$$L_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$
$$L_2 : \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

First, we set the right-hand sides of the two equations equal to each other:

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$

Now, we separate this equation into 3 independent equations, each relating to one component:

$$\begin{aligned} 1 + 1\lambda &= 0 + 2\mu \\ 2 + 0\lambda &= 2 + 1\mu \\ 3 + 1\lambda &= 2 - 1\mu \end{aligned}$$

Now, we solve any two of these equations. In this example, we'll pick the first and the last, but you can pick any two (try this, if you want!). From the first equation:

$$1 + \lambda = 2\mu \implies \lambda = 2\mu - 1$$

From the second equation:

$$3 + \lambda = 2 - \mu \implies \lambda = -\mu - 1$$

This tells us:

$$2\mu - 1 = -\mu - 1 \implies \mu = 0 \implies \lambda = -1$$

We substitute our calculated values of λ and μ into the unused equation, in this case, the second equation:

$$2 + 0(-1) = 2 = 2 + 1(0)$$

This proves that both sides of the equation are equal, so we can now substitute our calculated value of λ or μ into our equation for L_1 or L_2, respectively:

$$\mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + (-1) \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \implies \mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

This gives us the position vector of the point of intersection, with coordinates (0, 2, 2)

Problem Find, in slope-intercept form, the equation of a line which passes through the point (8, 4) and has normal vector $9\mathbf{i} + 4\mathbf{j}$

2. Mathematics

2.1 Theorems

2.1.1 Several equations

This is a theorem consisting of several equations.

Theorem 2.1 — Name of the theorem. In $E = \mathbb{R}^n$ all norms are equivalent. It has the properties:

$$\big|||\mathbf{x}|| - ||\mathbf{y}||\big| \leq ||\mathbf{x} - \mathbf{y}|| \tag{2.1}$$

$$||\sum_{i=1}^{n} \mathbf{x}_i|| \leq \sum_{i=1}^{n} ||\mathbf{x}_i|| \quad \text{where } n \text{ is a finite integer} \tag{2.2}$$

2.1.2 Single Line

This is a theorem consisting of just one line.

Theorem 2.2 A set $\mathscr{D}(G)$ in dense in $L^2(G)$, $|\cdot|_0$.

2.2 Definitions

A definition can be mathematical or it could define a concept.

Definition 2.1 — Definition name. Given a vector space E, a norm on E is an application, denoted $||\cdot||$, E in $\mathbb{R}^+ = [0, +\infty[$ such that:

$$||\mathbf{x}|| = 0 \;\Rightarrow\; \mathbf{x} = \mathbf{0} \tag{2.3}$$

$$||\lambda \mathbf{x}|| = |\lambda| \cdot ||\mathbf{x}|| \tag{2.4}$$

$$||\mathbf{x} + \mathbf{y}|| \leq ||\mathbf{x}|| + ||\mathbf{y}|| \tag{2.5}$$

2.3 Notations

■ **Notation 2.1** Given an open subset G of $\mathbb{R}^n$, the set of functions φ are:

1. Bounded support G;

2. Infinitely differentiable;

a vector space is denoted by $\mathscr{D}(G)$.

2.4 Remarks

This is an example of a remark.

R The concepts presented here are now in conventional employment in mathematics. Vector spaces are taken over the field $\mathbb{K} = \mathbb{R}$, however, established properties are easily extended to $\mathbb{K} = \mathbb{C}$.

2.5 Corollaries

Corollary 2.1 — Corollary name. The concepts presented here are now in conventional employment in mathematics. Vector spaces are taken over the field $\mathbb{K} = \mathbb{R}$, however, established properties are easily extended to $\mathbb{K} = \mathbb{C}$.

2.6 Propositions

2.6.1 Several equations

Proposition 2.1 — Proposition name. It has the properties:

$$\big|||\mathbf{x}|| - ||\mathbf{y}||\big| \leq ||\mathbf{x} - \mathbf{y}|| \tag{2.6}$$

$$||\sum_{i=1}^{n} \mathbf{x}_i|| \leq \sum_{i=1}^{n} ||\mathbf{x}_i|| \quad \text{where } n \text{ is a finite integer} \tag{2.7}$$

2.6.2 Single Line

Proposition 2.2 Let $f, g \in L^2(G)$; if $\forall \varphi \in \mathscr{D}(G)$, $(f, \varphi)_0 = (g, \varphi)_0$ then $f = g$.

2.7 Examples

2.7.1 Equation Example

■ **Example 2.1** Let $G = \{x \in \mathbb{R}^2 : |x| < 3\}$ and denoted by: $x^0 = (1, 1)$; consider the function:

$$f(x) = \begin{cases} \mathrm{e}^{|x|} & \text{si } |x - x^0| \leq 1/2 \\ 0 & \text{si } |x - x^0| > 1/2 \end{cases} \tag{2.8}$$

The function f has bounded support, we can take $A = \{x \in \mathbb{R}^2 : |x - x^0| \leq 1/2 + \varepsilon\}$ for all $\varepsilon \in]0; 5/2 - \sqrt{2}[$. ■

2.7.2 Text Example

■ **Example 2.2 — Example name.** Aliquam arcu turpis, ultrices sed luctus ac, vehicula id metus. Morbi eu feugiat velit, et tempus augue. Proin ac mattis tortor. Donec tincidunt, ante rhoncus luctus semper, arcu lorem lobortis justo, nec convallis ante quam quis lectus. Aenean tincidunt sodales massa, et hendrerit tellus mattis ac. Sed non pretium nibh. Donec cursus maximus luctus. Vivamus lobortis eros et massa porta porttitor. ■

2.8 Exercises

Worked Example This is a good place to ask a question to test learning progress or further cement ideas into students' minds. ■

2.9 Problems

Problem What is the average airspeed velocity of an unladen swallow?

2.10 Vocabulary

Define a word to improve a students' vocabulary.

■ **Vocabulary 2.1 — Word.** Definition of word.

3. Presenting Information and Results with a Long Chapter Title

3.1 Table

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Treatments	**Response 1**	**Response 2**
Treatment 1	0.0003262	0.562
Treatment 2	0.0015681	0.910
Treatment 3	0.0009271	0.296

Table 3.1: Table caption.

Referencing Table 3.1 in-text using its label.

3.2 Figure

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Treatments	**Response 1**	**Response 2**
Treatment 1	0.0003262	0.562
Treatment 2	0.0015681	0.910
Treatment 3	0.0009271	0.296

Table 3.2: Floating table.

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Figure 3.1: Figure caption.

Referencing Figure 3.1 in-text using its label.

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Figure 3.2: Floating figure.

Bibliography

Articles

Books

Index

Corollaries, 12

Definitions, 11

Examples, 12
    Equation, 12
    Text, 12
Exercises, 12

Figure, 15

Notations, 11

Problems, 13
Propositions, 12
    Several Equations, 12
    Single Line, 12

Remarks, 12

Table, 15
Theorems, 11
    Several Equations, 11
    Single Line, 11

Vocabulary, 13

A. Appendix Chapter Title

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A.1 Appendix Section Title

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B. Appendix Chapter Title

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B.1 Appendix Section Title

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